

COUNTEREXAMPLES TO THE HYPERCUBE MAXIMUM PROPAGATION-TIME CONJECTURE

ABSTRACT. Brimkov, Cameron, and Grubbs conjectured that the maximum propagation time of the hypercube Q_d , taken over its minimum zero forcing sets, is 2^{d-2} . We give explicit minimum zero forcing sets of Q_6 and Q_7 with propagation times 18 and 43, respectively. Hence $\text{PT}(Q_6) \geq 18 > 16$ and $\text{PT}(Q_7) \geq 43 > 32$, disproving the conjectured formula. Cameron and Pulaj reported the conjectured values for $2 \leq d \leq 5$, so Q_6 is the smallest-dimensional counterexample to this formula. We do not determine either exact maximum or address the full propagation-time-interval assertion in the original conjecture.

1. THE COUNTEREXAMPLES

We use the standard synchronous zero forcing convention of [1]. Identify the vertices of Q_d with $V_d = \{0, 1, \dots, 2^d - 1\}$ through their length- d binary expansions. Thus $x, y \in V_d$ are adjacent exactly when $x \oplus y$ is a power of two, where $\oplus$ denotes bitwise exclusive-or. For $C \subseteq V_d$, let

$$\Phi_d(C) = \{v \in V_d \setminus C : \text{there is } u \in C \text{ with } N(u) \setminus C = \{v\}\},$$

where $N(u)$ is the open neighborhood of u . Starting from $C_0 = S$, define

$$D_t = \Phi_d(C_{t-1}), \quad C_t = C_{t-1} \cup D_t \quad (t \geq 1).$$

Choosing one witness u for each $v \in D_t$ gives independent forces, since a vertex with a unique neighbor outside C_{t-1} cannot witness two distinct targets. Thus these are exactly the synchronous forcing layers. If $C_T = V_d$ and $C_{T-1} \neq V_d$, then $\text{pt}(Q_d, S) = T$. The parameter $\text{PT}(G)$ is the maximum of $\text{pt}(G, S)$ over all minimum zero forcing sets S of G .

Brimkov, Cameron, and Grubbs conjectured that

$$\text{PT}(Q_d) = 2^{d-2} \quad (d \geq 2)$$

as part of their Conjecture 4.4 [2]. Cameron and Pulaj restated this conjecture and reported its validity for $2 \leq d \leq 5$ [3, Conjecture 2.1 and Table 7].

Theorem 1. *Let*

$$S_6 = \{2, 5, 6, 7, 10, 13, 16, 17, 18, 20, 21, 22, 25, 28, 29, 30, 31, \\ 33, 35, 36, 38, 43, 44, 49, 50, 51, 56, 57, 58, 60, 62, 63\}$$

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and

$$\begin{aligned} S_7 = \{ & 0, 1, 2, 3, 4, 10, 11, 13, 14, 15, 21, 22, 24, 27, 30, 31, \\ & 34, 36, 37, 38, 39, 41, 45, 46, 48, 49, 50, 51, 55, 56, 57, 58, 63, \\ & 64, 65, 66, 67, 68, 69, 71, 74, 77, 79, 85, 88, 90, 93, 98, 100, 101, \\ & 102, 103, 108, 109, 112, 113, 115, 119, 120, 122, 123, 124, 126, 127 \}. \end{aligned}$$

Then S_6 and S_7 are minimum zero forcing sets and

$$\text{pt}(Q_6, S_6) = 18, \quad \text{pt}(Q_7, S_7) = 43.$$

Consequently,

$$\text{PT}(Q_6) \geq 18 > 2^{6-2}, \quad \text{PT}(Q_7) \geq 43 > 2^{7-2}.$$

In particular, the maximum-propagation-time assertion in [2, Conjecture 4.4] is false.

Proof. The known identity $Z(Q_d) = 2^{d-1}$ [4, Theorem 3.1] gives $Z(Q_6) = 32$ and $Z(Q_7) = 64$. The displayed sets have these respective cardinalities. It therefore suffices to verify that they force all vertices and to compute their propagation times. All calculations below use

$$N(u) = \{u \oplus 2^i : 0 \leq i < d\}$$

and exact set operations.

For S_6 , direct evaluation of the recurrence gives

$$\begin{aligned} (D_1, \dots, D_6) &= (\{61\}, \{52\}, \{4\}, \{14\}, \{26\}, \{19\}), \\ (D_7, \dots, D_{11}) &= (\{1\}, \{37\}, \{32\}, \{41\}, \{59\}), \\ (D_{12}, \dots, D_{15}) &= (\{27, 42, 55\}, \{11, 24, 47\}, \{8, 9, 12\}, \{0\}), \\ (D_{16}, D_{17}, D_{18}) &= (\{3, 40, 48\}, \{15, 23, 34, 45, 53\}, \{39, 46, 54\}). \end{aligned}$$

The listed layers are disjoint and their union is $V_6 \setminus S_6$. Hence $C_{18} = V_6$ and $D_{18} \neq \emptyset$, so $\text{pt}(Q_6, S_6) = 18$.

For S_7 , the exact layers are

$$\begin{aligned} (D_1, \dots, D_{10}) &= (\{5\}, \{7\}, \{47\}, \{35\}, \{19\}, \{59\}, \{43\}, \{111\}, \{99\}, \{33\}), \\ (D_{11}, \dots, D_{20}) &= (\{53\}, \{17\}, \{9\}, \{75\}, \{83\}, \{114\}, \{106\}, \{96\}, \{116\}, \{80\}), \\ (D_{21}, \dots, D_{30}) &= (\{72\}, \{78\}, \{95\}, \{125\}, \{105\}, \{40\}, \{60\}, \{92\}, \{89\}, \{29\}), \\ (D_{31}, \dots, D_{36}) &= (\{12, 23\}, \{6, 18, 20, 54, 87\}, \{70, 86\}, \{82\}, \{118\}, \{110, 117\}), \\ (D_{37}, \dots, D_{40}) &= (\{97, 107\}, \{32, 42\}, \{104\}, \{121\}), \\ (D_{41}, D_{42}, D_{43}) &= (\{61, 73, 81, 91\}, \{25, 44, 52, 62, 76, 84, 94\}, \{8, 16, 26, 28\}). \end{aligned}$$

These layers are disjoint and their union is $V_7 \setminus S_7$. Thus $C_{43} = V_7$ and $D_{43} \neq \emptyset$, proving $\text{pt}(Q_7, S_7) = 43$.

Both sets are therefore zero forcing sets of minimum cardinality. The bounds on $\text{PT}(Q_6)$ and $\text{PT}(Q_7)$ follow from its definition. $\square$

Remark. The Q_6 certificate alone disproves the formula $\text{PT}(Q_d) = 2^{d-2}$. The theorem gives only lower bounds on $\text{PT}(Q_6)$ and $\text{PT}(Q_7)$; it does not determine either exact maximum. It also does not decide whether Q_6 or Q_7 has a full realized propagation-time interval, the separate interval assertion in [2, Conjecture 4.4].

REFERENCES

- [1] L. Hogben, M. Huynh, N. Kingsley, S. Meyer, S. Walker, and M. Young, *Propagation time for zero forcing on a graph*, Discrete Appl. Math. **160** (2012), no. 13–14, 1994–2005, doi:10.1016/j.dam.2012.04.003.
- [2] B. Brimkov, T. R. Cameron, and O. Grubbs, *On the forts and related parameters of the hypercube graph*, arXiv:2507.10826v1 (2025).
- [3] T. R. Cameron and J. Pulaj, *IP models for minimum zero forcing sets, forts, and related graph parameters*, arXiv:2508.07293v1 (2025).
- [4] AIM Minimum Rank–Special Graphs Work Group, *Zero forcing sets and the minimum rank of graphs*, Linear Algebra Appl. **428** (2008), no. 7, 1628–1648, doi:10.1016/j.laa.2007.10.009.